Executive Report -- Adult Income Classifier

Business goal: predict whether an individual's annual income exceeds $50K, to support targeting/prioritization decisions that depend on likely income level.

Data & class imbalance

UCI/OpenML Adult dataset (48842 rows, 14 raw features). Positive class (income >$50K) is 23.9% of the data (3.18:1 negative:positive).

Final model & imbalance strategy

Final model: **HistGradientBoostingClassifier**. Final imbalance-handling strategy: **class_weight=balanced**, selected by mean cross-validation F1 on X_train (not by test-set performance). Final production threshold: **0.63**, selected on a dedicated threshold-validation split before the model was refit on all of X_train.

Key hold-out metrics (test set, frozen threshold)

Threshold	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
0.63	0.8561	0.6728	0.7764	0.7209	0.9255	0.8254

Ensemble comparison (Task 1, test set, 0.5 cutoff)

Model	F1	ROC AUC	PR AUC	Inference (ms/row)
Stacking (RF+HistGB+XGB)	0.7152	0.9272	0.83	0.0182
Day-4 Single Gradient Boosting (baseline)	0.7101	0.9263	0.8269	0.0067
HistGradientBoosting	0.7101	0.9263	0.8269	0.0068
XGBoost	0.7077	0.927	0.8288	0.004
Random Forest	0.6736	0.9151	0.8028	0.0129

Interpretability -- top features

Top contributing features by mean |SHAP value| (transformed/one-hot feature space): marital-status_Married-civ-spouse, age, capital-gain, education_hours, education-num, marital-status_Never-married, sex_Female, capital-loss.

Local explanation summary

True Positive (row 13): probability 0.7322 vs threshold 0.63. Top contributing features: marital-status_Married-civ-spouse (pushed the prediction toward >$50K), age (pushed the prediction toward >$50K), education_hours (pushed the prediction toward >$50K).

False Positive (row 7): probability 0.7039 vs threshold 0.63. Top contributing features: marital-status_Married-civ-spouse (pushed the prediction toward >$50K), race_Asian-Pac-Islander (pushed the prediction toward >$50K), relationship_Wife (pushed the prediction toward >$50K).

False Negative (row 3): probability 0.2659 vs threshold 0.63. Top contributing features: age (pushed the prediction toward >$50K), marital-status_Married-civ-spouse (pushed the prediction toward <=$50K), education_hours (pushed the prediction toward <=$50K).

Fairness observations (numerical gaps)

By sex: F1 gap 0.0343, recall gap 0.1259, predicted-positive-rate gap 0.2380 (worst group: Female).
By race: F1 gap 0.1000, recall gap 0.2865, predicted-positive-rate gap 0.2566 (worst group: Amer-Indian-Eskimo). These are associational

gaps, not causal claims; see the notebook's Disparities & Mitigations section for proposed mitigations.

Deployment artifact

capstone_income_pipeline.joblib (pipeline + frozen threshold + imbalance_approach + raw_input_columns + sklearn_version + random_state), served via inference.py (dict / list[dict] / DataFrame in, probability + predicted class + threshold + top-3 contributing features out).

Recommended next steps

1) Run an A/B test or controlled pilot: route a held-out slice of live traffic through the new model alongside the current process, compare business-relevant outcomes (not just F1) before full rollout. 2) Re-check the winning imbalance strategy on the stacking/XGBoost architectures from Task 1 to confirm the choice generalizes. 3) Stand up the monitoring checklist (data drift, score drift, predicted-positive rate, rolling F1/recall, fairness gaps) before go-live, with the alert thresholds documented in the notebook. 4) Schedule quarterly retraining, re-running the full model/imbalance/fairness pipeline, with an event-triggered retrain path for confirmed drift.